

Problem 1: The surface of a red giant

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Take a giant with $M = 1 M_{\odot}$, $R = 50 R_{\odot}$, $T_{\text{eff}} = 4000 \text{ K}$ and $\mu = 1.3$.

- (a) From hydrostatic equilibrium and the ideal gas law, derive $H_P = kT/\mu m_H g$ for a thin atmosphere. Say where each assumption enters. (10)

Starting from hydrostatic equilibrium, we know $\frac{dP}{dr} = -pg$

↳ This states that the inward pull of gravity is balanced by the outward pressure gradient for the star.

To get this relation, we started with the momentum equation $\rho \frac{du}{dt} + \rho u \frac{du}{dr} = -\frac{\partial P}{\partial r} - \frac{Gm(r)\rho}{r^2}$ and assume the gas is in hydrostatic equilibrium. This means that the gas (which we are treating as a fluid) has no bulk radial motion (the gas isn't rising or sinking). Consequently, the velocity u is zero, and so is its time derivative.

$$\therefore \rho \frac{du}{dt} = 0, \rho u \frac{du}{dr} = 0 \Rightarrow \therefore 0 = -\frac{\partial P}{\partial r} - \frac{Gm(r)\rho}{r^2} \Rightarrow \therefore \frac{\partial P}{\partial r} = -\frac{Gm(r)\rho}{r^2}$$

By assuming the star has a thin atmosphere, we can simplify this expression.

If the atmosphere is thin relative to the stellar radius, we can approximate any radial position in the atmosphere to be the stellar radius $\Rightarrow r \approx R \Rightarrow m(r) \approx M$.

Consequently, hydrostatic equilibrium then gives us $\frac{\partial P}{\partial r} = -\frac{GM\rho}{R^2}$

Recall that gravitational acceleration is given by $g = \frac{GM}{R^2} \Rightarrow \therefore \frac{\partial P}{\partial r} = -pg$

If we look at the definition of the pressure scale height $H_P = -\left(\frac{d \ln(P)}{dr}\right)^{-1}$

We can rewrite the derivative using chain rule: $\frac{d \ln(P)}{dr} = \frac{1}{P} \frac{dP}{dr}$

$\therefore$ The definition of pressure scale height becomes $H_P = \frac{-P}{\left(\frac{dP}{dr}\right)}$

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We can substitute the result of hydrostatic equilibrium and the thin atmosphere approximation into the pressure scale height definition:

$$\Rightarrow H_p = \frac{P}{(-\rho g)} \Rightarrow \boxed{H_p = \frac{P}{\rho g}}$$

Assuming the gas in the stellar atmosphere can be approximated as an ideal gas, where $P = \frac{\rho k T}{\mu m_H}$, we can substitute the pressure for the ideal gas law:

$$H_p = \left(\frac{\rho k T}{\mu m_H} \right) \left(\frac{1}{\rho g} \right) \Rightarrow \boxed{\therefore H_p = \frac{k T}{\mu m_H g}}$$

Which is the form we have been looking for. This gives us the distance over which the pressure in the stellar atmosphere changes by a factor of e .

(b) Evaluate H_p and H_p/R for the giant and compare with the Sun. Account for the ratio of the two H_p/R values by scaling, not by recomputing. (10)

Variables

- $G = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$
- $M_\odot = 1.99 \times 10^{33} \text{ g}$
- $R_\odot = 6.96 \times 10^9 \text{ cm}$
- $T = 4000 \text{ K}$
- $k = 1.38 \times 10^{-16} \text{ erg K}^{-1}$
- $\mu = 1.3$
- $m_H = 1.67 \times 10^{-24} \text{ g}$

To calculate the scale height, we first must find the gravitational acceleration for a star of mass $M = 1M_\odot$ and radius $R = 50R_\odot$:

$$g = \frac{GM}{R} \Rightarrow g = \frac{G M_\odot}{50 R_\odot} \Rightarrow \boxed{g = 11.0 \text{ cm/s}^2}$$

Next we can calculate the scale height:

$$H_p = \frac{kT}{\mu m_H g} \Rightarrow \boxed{\therefore H_p = 2.3 \times 10^{10} \text{ cm}}$$

$\therefore$ A star with mass $M = 1M_\odot$ and $R = 50R_\odot$ has a pressure scale height of $2.3 \times 10^{10} \text{ cm}$. This means that over a distance of $2.3 \times 10^{10} \text{ cm}$, the pressure changes by a factor of e .

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Finally, the scale height compared to its radius is given by $\frac{H_p}{R}$.

$$\text{For the giant, we have } \frac{H_p}{R} = \frac{2.3 \times 10^{10} \text{ cm}}{50 R_o} \Rightarrow \therefore \frac{H_p}{R} = 0.0066 \text{ or } 0.67\%$$

$\therefore$ for the giant, 1 scale height only accounts for 0.67% of its radius.

To compare the scale height-to-radius ratio of the giant to our sun, we will simply take a ratio:

$$\frac{\left(\frac{H_p}{R}\right)_G}{\left(\frac{H_p}{R}\right)_O} = \frac{\left(\frac{k T_G}{\mu_G m_H g_G}\right)}{\left(\frac{k T_O}{\mu_O m_H g_O}\right)} \Rightarrow \frac{\left(\frac{H_p}{R}\right)_G}{\left(\frac{H_p}{R}\right)_O} = \left(\frac{T_G}{T_O}\right) \left(\frac{\mu_O}{\mu_G}\right) \left(\frac{g_O}{g_G}\right) \left(\frac{R_O}{R_G}\right)$$

$$\text{Recall } g = \frac{GM}{r^2} \Rightarrow g_G = \frac{GM_O}{(50 R_O)^2}, g_O = \frac{GM_O}{R_O^2} \Rightarrow \left(\frac{g_O}{g_G}\right) \left(\frac{R_O}{R_G}\right) = (50)^2 \left(\frac{1}{50}\right) = 50$$

$$\therefore \frac{\left(\frac{H_p}{R}\right)_G}{\left(\frac{H_p}{R}\right)_O} = 50 \left(\frac{T_G}{T_O}\right) \left(\frac{\mu_O}{\mu_G}\right), \text{ but } \mu_O = 1.3 = \mu_G \Rightarrow \frac{\left(\frac{H_p}{R}\right)_G}{\left(\frac{H_p}{R}\right)_O} = 50 \frac{T_G}{T_O}$$

$$\text{Note: } T_G = 4000 \text{ K}, T_O = 5770 \text{ K} \Rightarrow \frac{T_G}{T_O} \approx 0.69$$

$$\therefore \frac{\left(\frac{H_p}{R}\right)_G}{\left(\frac{H_p}{R}\right)_O} \approx 35 \Rightarrow \text{This means that the scale height-to-radius ratio for the giant is 35 times larger than that same ratio for the Sun.}$$

The scale height of the Giant is a larger fraction of its radius than the sun's scale height is to its own radius.

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- (c) The giant's central pressure exceeds its photospheric pressure by fifteen orders of magnitude. How many pressure scale heights is it tall, and why do you not need a stellar model to say so? (5)

We are told that $\frac{P_c}{P_p} \approx 10^{15}$

We know that the scale height H_p is the distance over which the pressure changes by a factor of e . This means that e multiplied by itself N times should give us 10^{15} .

$\therefore$ We want to figure out what power N of e gives us $10^{15} \Rightarrow e^N = 10^{15}$
Here N is the number of scale heights.

$$\Rightarrow \ln(e^N) = \ln(10^{15}) \Rightarrow N = \ln(10^{15}) \Rightarrow N = 15 \ln(10) \Rightarrow \therefore N \approx 35$$

This means the central pressure is 35 scale heights tall.

Problem 2: Timescales of a massive star

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Take a main-sequence star with $M = 10 M_{\odot}$, $R = 4 R_{\odot}$ and $L = 6000 L_{\odot}$.

- (a) From the virial theorem for an ideal monatomic gas, show that $E = -U$. The star contracts and releases gravitational energy $|\Delta\Omega|$: what fraction is radiated, and where does the rest go? (8)

We know from virial theorem that $2U + \Omega = 0$. We also know that the total energy of the system is given by $E = \Omega + U$.

$$\therefore \Omega = -2U \Rightarrow E = (-2U) + U \Rightarrow \boxed{\therefore E = -U}$$

This result means that as the star radiates away energy, the internal energy increases, causing the star to get hotter. By virial theorem, this increase in internal energy is supplied by a decrease in gravitational potential energy as the star contracts.

When a star contracts such that its gravitational energy changes by $\Delta\Omega$, by virial theorem, the internal (thermal) energy must also change.

$$\therefore 2\Delta U + \Delta\Omega = 0 \Rightarrow \boxed{\Delta U = -\frac{1}{2}\Delta\Omega}$$

$\therefore$ As the star contracts, the gravitational energy is negative ($\Delta\Omega < 0$), so half of the gravitational energy goes into the internal (thermal) energy, heating up the star.

If we then consider the change in the total energy ΔE , we have $\Delta E = \Delta U + \Delta\Omega$, but by virial theorem $\Delta U = -\frac{1}{2}\Delta\Omega$.

$$\Rightarrow \Delta E = -\frac{1}{2}\Delta\Omega + \Delta\Omega \Rightarrow \boxed{\therefore \Delta E = \frac{1}{2}\Delta\Omega} \Rightarrow \text{Since } \Delta\Omega < 0 \Rightarrow \Delta E < 0$$

$\therefore$ Half of the energy is radiated away.

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- (b) Evaluate the dynamical, thermal and nuclear timescales. For the nuclear one, assume a tenth of the mass burns hydrogen at an efficiency of 0.007. (12)

From the lecture 3 slides, we have the following:

Dynamical Timescale: $t_{\text{dyn}} \approx \sqrt{\frac{R^3}{GM}}$

Thermal Timescale: $t_{\text{KH}} \approx \frac{GM^2}{RL}$

Nuclear Timescale: $t_{\text{nuc}} \approx \frac{(0.007)(0.1)Mc^2}{L}$

Note: 0.007 is the hydrogen burning efficiency, and 0.1 of the mass is burned.

We are given $M = 10M_{\odot}$, $R = 4R_{\odot}$, $L = 6000L_{\odot}$.

Note: $M_{\odot} = 1.99 \times 10^{30} \text{ kg}$, $R_{\odot} = 6.957 \times 10^8 \text{ m}$, $L_{\odot} = 3.828 \times 10^{26} \text{ J/s}$, $G = 6.67 \times 10^{-11} \text{ m}^3/\text{kg s}^2$

① Dynamical Timescale:

$$t_{\text{dyn}} \approx \sqrt{\frac{(4R_{\odot})^3}{G(10M_{\odot})}} \Rightarrow \boxed{t_{\text{dyn}} \approx 4029 \text{ s}} \quad \text{or} \quad \boxed{t_{\text{dyn}} \approx 67.15 \text{ min}}$$

The dynamical timescale, which is the time it would take the star to collapse under its own weight, is $t_{\text{dyn}} \approx 4029 \text{ s}$.

② Thermal Timescale: $t_{\text{KH}} \approx \frac{G(10M_{\odot})^2}{(4R_{\odot})(6000L_{\odot})}$

$$\Rightarrow \boxed{t_{\text{KH}} \approx 4.13 \times 10^{12} \text{ s} \quad \text{or} \quad t_{\text{KH}} \approx 1.31 \times 10^5 \text{ yr}}$$

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The thermal timescale is the time under which the star would take to radiate away all its energy assuming there is no nuclear fusion. For this particular star, we found the thermal timescale to be $t_{KH} \approx 1.31 \times 10^5 \text{ yr}$.

③ Nuclear Timescale: $t_{nuc} \approx \frac{(0.007)(0.1)(10M_{\odot})c^2}{(6000L_{\odot})}$

Here c is the speed of light: $c = 3 \times 10^8 \text{ m/s}$ $\therefore t_{nuc} \approx 5.46 \times 10^{14} \text{ s}$ or $t_{nuc} \approx 1.73 \times 10^7 \text{ yr}$

The nuclear timescale is how long a star can continue to radiate at its current luminosity given the energy is released by nuclear fusion. For this star, the nuclear timescale is $1.73 \times 10^7 \text{ yr}$.

As expected, for this star we find $t_{dyn} < t_{KH} < t_{nuc}$.

(c) Compare each with its solar value using the scalings, and say which of the three ratios differs most from the Sun's and why. (5)

Dynamical Timescale: $\frac{(t_{dyn})_s}{(t_{dyn})_{\odot}} = \left[\frac{\left(\frac{(4R_{\odot})^3}{G(10M_{\odot})} \right)}{\left(\frac{(R_{\odot})^3}{G(M_{\odot})} \right)} \right]^{1/2} \Rightarrow \frac{(t_{dyn})_s}{(t_{dyn})_{\odot}} = \sqrt{\frac{64}{10}}$

$\therefore \frac{(t_{dyn})_s}{(t_{dyn})_{\odot}} = 2.53 \Rightarrow$ The dynamical timescale is 2.53 times longer than that of the Sun.

Thermal Timescale: $\frac{(t_{KH})_s}{(t_{KH})_{\odot}} = \frac{\left(\frac{G(10M_{\odot})^2}{(4R_{\odot})(6000L_{\odot})} \right)}{\left(\frac{G(M_{\odot})^2}{(R_{\odot})(L_{\odot})} \right)} \Rightarrow \frac{(t_{KH})_s}{(t_{KH})_{\odot}} = \frac{100}{24000}$

$\therefore \frac{(t_{KH})_s}{(t_{KH})_{\odot}} = 4.17 \times 10^{-3} \Rightarrow \frac{(t_{KH})_{\odot}}{(t_{KH})_s} = 240 \Rightarrow$ The thermal timescale of the Sun is 240 times longer than this star.

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Nuclear Timescale: Assuming the same fraction of mass is burned and hydrogen is burned at the same efficiency in the sun, we have:

$$\frac{(t_{\text{nuc}})_s}{(t_{\text{nuc}})_\odot} = \frac{\left(\frac{(0.007)(0.1)(10M_\odot)c^2}{(6000L_\odot)}\right)}{\left(\frac{(0.007)(0.1)(M_\odot)c^2}{(L_\odot)}\right)} \Rightarrow \frac{(t_{\text{nuc}})_s}{(t_{\text{nuc}})_\odot} = \frac{10}{6000} \Rightarrow \therefore \frac{(t_{\text{nuc}})_s}{(t_{\text{nuc}})_\odot} = 1.67 \times 10^{-2} \text{ or } \frac{(t_{\text{nuc}})_\odot}{(t_{\text{nuc}})_s} = 600$$

The nuclear timescale of the sun is 60 times larger than the nuclear timescale of this star.

Overall, the closest in timescales between the two stars was the dynamical timescale.

The ratio between the nuclear timescales differed the most, with the sun's nuclear timescale being 600 times longer than the star. This is because even though the $10M_\odot$ star has more fuel, it is 6000 times more luminous than the sun. It burns through its fuel much faster than the sun because it is so much more luminous, outweighing the increased fuel supply.

Problem 3: Composition and ionization

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- (a) A helium-burning core has mass fractions $Y = 0.5$, $^{12}\text{C} = 0.3$ and $^{16}\text{O} = 0.2$, fully ionized. Compute μ and μ_e exactly, species by species. Then compute μ from $1/\mu \approx 2X + \frac{3}{4}Y + \frac{1}{2}Z$ and say which approximation causes the difference. (12)

We know generally that $\frac{1}{\mu} = \frac{n m_H}{\rho}$.

Helium: Consider the number density of Helium, which will be the sum of the number density of free electrons and nuclei $\Rightarrow n_{\text{He}} = n_{\text{nuc,He}} + n_{e,\text{He}}$

For fully ionized He, we have 3 free particles (2e & 1 nuclei), and the nucleus has mass $4m_H$.

$$\therefore \frac{n_{e,\text{He}}}{n_{\text{nuc,He}}} = 2 \Rightarrow n_{\text{He}} = (2n_{\text{nuc,He}}) + n_{\text{nuc,He}}$$

$$\Rightarrow n_{\text{He}} = 3n_{\text{nuc,He}} \Rightarrow \therefore n_{\text{He}} = \frac{3\rho_{\text{He}}}{4m_H}, \text{ but } \rho_{\text{He}} = Y\rho \Rightarrow \boxed{\therefore n_{\text{He}} = \left(\frac{3}{4}Y\right)\frac{\rho}{m_H}}$$

$$\text{Also note that since } n_{e,\text{He}} = 2n_{\text{nuc,He}} \Rightarrow n_{e,\text{He}} = \frac{\rho_{\text{He}}}{2m_H} \Rightarrow \boxed{\therefore n_{e,\text{He}} = \left(\frac{1}{2}Y\right)\frac{\rho}{m_H}}$$

Carbon 12: Generally, $n_{\text{nuc},i} = \frac{\rho_i}{A_i m_H}$

Carbon 12 has an atomic mass number of $A_c = 12$, and when fully ionized contributes 6 free electrons for every nuclei

$$\therefore n_{\text{nuc},c} = \frac{\rho_c}{12m_H}, \quad n_{e,c} = \frac{6\rho_c}{12m_H} \Rightarrow \therefore n_c = \frac{7\rho_c}{12m_H}, \text{ but } \rho_c = Z_c \rho$$

$$\Rightarrow \boxed{\therefore n_c = \left(\frac{7}{12}Z_c\right)\frac{\rho}{m_H}}, \text{ also note that } n_{e,c} = \frac{6\rho_c}{12m_H} \Rightarrow n_{e,c} = \frac{\rho_c}{2m_H} \Rightarrow \boxed{n_{e,c} = \left(\frac{1}{2}Z_c\right)\frac{\rho}{m_H}}$$

Oxygen 16: Has an atomic mass number $A_o = 16$ and when fully ionized has 8 free electrons.

$$\therefore n_{\text{nuc},o} = \frac{\rho_o}{16m_H}, \quad n_{e,o} = \frac{8\rho_o}{16m_H} \Rightarrow \therefore n_o = \frac{9\rho_o}{16m_H}, \text{ but } \rho_o = Z_o \rho \Rightarrow \boxed{\therefore n_o = \left(\frac{9}{16}Z_o\right)\frac{\rho}{m_H}}$$

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Note that since $n_{e,o} = \frac{8\rho_o}{16m_H}$ and $\rho_c = Z_c \rho$, $n_{e,o} = \left(\frac{1}{2}Z_o\right)\frac{\rho}{m_H}$

finally we can bring these results together to find μ & μ_e :

$$n = n_{He} + n_c + n_o \Rightarrow \frac{\rho}{\mu m_H} = \frac{\rho}{m_H} \left(\frac{3}{4}Y\right) + \frac{\rho}{m_H} \left(\frac{7}{12}Z_c\right) + \frac{\rho}{m_H} \left(\frac{9}{16}Z_o\right)$$

$$\therefore \frac{1}{\mu} = \frac{3}{4}Y + \frac{7}{12}Z_c + \frac{9}{16}Z_o$$

for $Y=0.5$, $Z_c=0.3$, $Z_o=0.2$, we have $\frac{1}{\mu} = 0.66 \Rightarrow \mu = 1.51$

We can also find μ_e the same way:

$$n_e = n_{e,He} + n_{e,c} + n_{e,o} \Rightarrow \frac{\rho}{\mu_e m_H} = \frac{\rho}{m_H} \left(\frac{1}{2}Y\right) + \frac{\rho}{m_H} \left(\frac{1}{2}Z_c\right) + \frac{\rho}{m_H} \left(\frac{1}{2}Z_o\right)$$

$$\Rightarrow \therefore \frac{1}{\mu_e} = \frac{1}{2}[Y + Z_c + Z_o] \Rightarrow \therefore \frac{1}{\mu_e} = \frac{1}{2} \Rightarrow \mu_e = 2$$

If we only use the approximation $\frac{1}{\mu} \approx 2X + \frac{3}{4}Y + \frac{1}{2}Z$,

We have $\frac{1}{\mu} \approx \frac{1}{2}X + \frac{3}{4}Y + \frac{1}{2}Z_c + \frac{1}{2}Z_o$, but $X=0$

$$\Rightarrow \therefore \frac{1}{\mu} \approx \frac{3}{4}Y + \frac{1}{2}Z_c + \frac{1}{2}Z_o \Rightarrow \therefore \frac{1}{\mu} \approx 0.63 \text{ or } \mu \approx 1.6$$

The approximation overestimates the mean molecular weight. This is because the approximation neglects the contribution of the nuclei, only accounting for the electron contribution.

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- (b) For pure hydrogen with total nucleus density $n = 10^{15} \text{ cm}^{-3}$, find the temperature at which the gas is half ionized. Compare it with χ/k and explain the ratio from the structure of the Saha equation. Then say how the answer moves if n falls by a factor of 1000, and check the direction against the equation. (13)

The Saha equation is the following: $\frac{n_e n_p}{n_H} = \frac{2 g_{e1}}{g_i} \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_i/kT}$

We can use the total nucleus density $n = 10^{15} \text{ cm}^{-3}$ and the fact that half of the gas is ionized.

We know that the total number density is the sum of the hydrogen and proton densities (ionized hydrogen is just a proton) $\Rightarrow n = n_H + n_p$

Since half the gas is ionized, we have $n_p = \frac{1}{2} n$. The fraction of ionization is then given by $x = \frac{n_p}{n}$, but $\frac{n_p}{n} = \frac{1}{2} \Rightarrow \therefore x = \frac{1}{2}$.

For ionized hydrogen $n_e = n_p$, and since $n_p = \frac{1}{2} n$, $n_e = n_p = \frac{1}{2} n$. Also, since $n = n_H + n_p$ we then have $n_H = \frac{1}{2} n$.

The statistical weight for ionized hydrogen is $g_{i1} = 1$ since a bare proton can only exist in one state. The electron in neutral hydrogen can be in two possible spin states so $g_i = 2$.

$$\therefore \frac{n_e n_p}{n_H} = \frac{(\frac{1}{2}n)(\frac{1}{2}n)}{(\frac{1}{2}n)} \Rightarrow \frac{n_e n_p}{n_H} = \frac{1}{2} n \quad \& \quad \frac{g_{i1}}{g_i} = \frac{1}{2}$$

The Saha equation then becomes $\frac{1}{2} n = \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-\chi_i/kT}$

Note:

$$\begin{aligned} \cdot n &= 10^{15} \text{ cm}^{-3} & \cdot k &= 1.381 \times 10^{-16} \text{ erg K}^{-1} & \cdot \chi_H &= 13.6 \text{ eV} = 2.179 \times 10^{-11} \text{ erg} \\ \cdot m_e &= 9.109 \times 10^{-28} \text{ g} & \cdot h &= 6.626 \times 10^{-27} \text{ erg s} \end{aligned}$$

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Subbing in those constants gives

$$(5.0 \times 10^{14} \text{ cm}^{-3}) = (2.41 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-3/2}) T^{3/2} e^{-(1.58 \times 10^5 \text{ K})/T}$$

$$\Rightarrow \therefore 0.207 = T^{3/2} e^{-(1.58 \times 10^5)/T} \Rightarrow 0.207 e^{(1.58 \times 10^5)/T} = T^{3/2}$$

We cannot solve this equation analytically but we can solve this numerically by finding the point of intersection between the left and right sides of the equation.

$$\text{Let } f(T) = 0.207 e^{(1.58 \times 10^5)/T}, g(T) = T^{3/2} \Rightarrow \text{finding } T \text{ for } f(T) = g(T)$$

This intersection occurs at $T \approx 1.02 \times 10^4 \text{ K}$.

$\therefore$ The hydrogen gas is half ionized at approximately $T \approx 1.02 \times 10^4 \text{ K}$

We know $\frac{\chi_H}{k} = 1.58 \times 10^5 \text{ K}$, and can compare the half ionization temperature to it by a ratio.

$$\Rightarrow \frac{T}{(\frac{\chi_H}{k})} \approx 0.065 \Rightarrow \boxed{\frac{(\frac{\chi_H}{k})}{T} \approx 16}$$

This means that the half ionization temperature is 16 times smaller than $\frac{\chi}{k}$, so the gas is half ionized even though $kT < \chi$.

The $e^{-\chi/kT}$ term is suppressing the ionization since $kT < \chi$ in this case, but the $(\frac{2\pi m_e kT}{h^2})^{3/2}$ term is very large. This is because this term favours ionization since a free electron has access to more translational states than a bound electron. The exponential term is the cost for ionizing hydrogen but the other term is the "reward" for ionization, being an increase in entropy by freeing electrons.

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To see what happens when n falls by a factor of 1000 ($n' = 10^{12} \text{ cm}^{-3}$), we have to sub in n' for n in the saha equation and numerically solve for T again.

$$\frac{1}{2} n' = \left(\frac{2\pi m_e kT}{h^2} \right)^{3/2} e^{-X_H/T} \Rightarrow 5 \times 10^{11} \text{ cm}^{-3} = 2.41 \times 10^{15} T^{3/2} e^{-1.58 \times 10^5 / T}$$

Numerically we solve this and find that when the density is decreased by a factor of 1000, the temperature at half ionization drops to $T' \approx 7.24 \times 10^3 \text{ K}$.

$\therefore$ When the density drops by a factor of 1000, the temperature drops from 10200 K to 7240 K. Since $n \propto T^{3/2} e^{-X_H/T}$, a decrease in the number density must decrease the temperature, which is exactly what we find.

Problem 4: The Heaviest Stars

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- (a) Evaluate the Eddington luminosity per solar mass for electron scattering, for $X = 0.7$ and for pure helium. (8)

From the slides, the Eddington Luminosity is given by $L_{\text{Edd}} = \frac{4\pi c G M}{k}$

where $k = 0.2(1+X)$ for electron scattering. The Eddington Luminosity per solar mass is then given by

$$\frac{L_{\text{Edd}}}{M_{\odot}} = \frac{4\pi c G M}{k M_{\odot}}$$

for $X=0.7$, we have $k=0.34 \text{ cm}^2 \text{ g}^{-1}$.

for $G = 6.674 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$, $c = 3.00 \times 10^{10} \text{ cm s}^{-1}$, $M = M_{\odot} = 1.99 \times 10^{33} \text{ g}$, we have

$$\frac{L_{\text{Edd,H}}}{M_{\odot}} = 1.47 \times 10^{38} \text{ erg/s } M_{\odot}$$

for pure helium $X=0$, so $k=0.20 \text{ cm}^2 \text{ g}^{-1}$, $\therefore \frac{L_{\text{Edd,He}}}{M_{\odot}} = 2.50 \times 10^{38} \text{ erg/s } M_{\odot}$

We find that the Eddington luminosity per solar mass is larger for pure helium.

- (b) Suppose main-sequence stars obeyed $L/L_{\odot} = (M/M_{\odot})^3$ at all masses. Find the mass at which a hydrogen-rich star would reach its Eddington luminosity. (10)

We want to find M for when $L = L_{\text{Edd}}$. We know $L_{\text{Edd}} = \frac{4\pi c G M}{k}$,

$$\Rightarrow \frac{4\pi c G M}{k L_{\odot}} = \left(\frac{M}{M_{\odot}}\right)^3 \Rightarrow M^2 = \frac{4\pi c G M_{\odot}^3}{k L_{\odot}} \Rightarrow \therefore M_{\text{Edd}} = \sqrt{\frac{4\pi c G M_{\odot}^3}{k L_{\odot}}}$$

for a hydrogen-rich star, assume that $X=0.7 \Rightarrow \therefore k=0.34 \text{ cm}^2 \text{ g}^{-1}$

Note: $L_{\odot} = 3.828 \times 10^{33} \text{ erg s}^{-1} \Rightarrow \therefore M_{\text{Edd}} = 3.90 \times 10^{35} \text{ g}$ or $M_{\text{Edd}} = 196 M_{\odot}$

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i. Assuming all main sequence stars obey $\frac{L}{L_\odot} = \left(\frac{M}{M_\odot}\right)^3$ at all masses, a hydrogen rich star ($X=0.7$) would need a mass of $196 M_\odot$ to reach its Eddington Luminosity.

(c) Envelope opacities in massive stars can exceed electron scattering several-fold. Take $\kappa = 1 \text{ cm}^2 \text{ g}^{-1}$, redo (b), and say what the result implies for how massive a star can be. Real luminosities rise more slowly than M^3 at high mass: which way does that move your answer? (7)

If $\kappa = 1 \text{ cm}^2 \text{ g}^{-1}$, using $M = \sqrt{\frac{4\pi c G M_\odot^3}{\kappa L_\odot}}$, we find

$$M_{\text{Edd}} = 2.28 \times 10^{35} \text{ g} \text{ or } M_{\text{Edd}} = 114 M_\odot$$

Larger opacity lowers the Eddington Luminosity, so by proportionality, $L \propto M^3$, this limit is reached at a lower mass. Knowing that real luminosities rise more slowly at high mass than M^3 , implies that we are underestimating the masses of large stars at the Eddington limit.